IoT-Enabled Smart Bus Route Monitoring and Revenue Management System

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Public bus transit systems in developing economies face acute structural and market failures, characterized by severe off-peak service abandonment, manual schedule administration, undetected route deviations, aggressive driving behavior, and flat-rate permit fees divorced from actual operational delivery. In Sri Lanka, despite statutory initiatives such as the National Transport Commission's (NTC) Nisi Seriya night bus service, private operators systematically curtail late-night (9:00 PM to 5:00 AM) and low-density operations due to economic deficits and the absence of objective physical verification mechanisms. This thesis presents Bus Sync, an end-to-end, IoT-enabled intelligent transportation and economic governance framework integrating vehicle telemetry, map-matching verification, dynamic policy-based scheduling, distance-based revenue management, and real-time driver safety feedback into a unified closed loop.

Onboard ESP32 edge units equipped with GPS and 6-axis IMU sensors continuously transmit spatial and kinetic telemetry over TLS-encrypted MQTT to an Azure IoT Hub gateway. A centralized Spring Boot backend engine executes point-to-segment orthogonal map-matching for corridor compliance, classifies operations into Peak, Normal, and Night time slots, automates distance-based permit fee calculations with verified night service credits, and detects dangerous driving maneuvers (over speeding, harsh braking, sharp turning, and abnormal crawling). Empirical evaluation over 100 test tracking samples, 500 trip records, and 120 controlled driver maneuver tests demonstrates a 4.2 m average GPS accuracy, 94.95% route-deviation detection, 96.2% time-slot classification accuracy, 0.5-minute delay estimation MAE, sub-3-second standby vehicle dispatching, 99.1% distance logging agreement, 100% permit fee calculation correctness, and 92.6% aggregate driver behavior detection accuracy. By tying financial permit liabilities and subsidies directly to verified physical movement, the proposed framework eliminates the off-peak economic deficit and establishes an accountable, data-driven transit paradigm.

Keywords: Intelligent Transportation Systems, Internet of Things, Driver Monitoring, Policy-Based Scheduling, Revenue Management, Map Matching, Smart Cities.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AVL	Automatic Vehicle Location
DSRM	Design Science Research Methodology
ESP32	Espressif Systems 32-bit Microcontroller with Wi-Fi & Bluetooth
GPRS	General Packet Radio Service (2G/3G Cellular Telemetry)
GPS	Global Positioning System
IMU	Inertial Measurement Unit (Accelerometer and Gyroscope)
IoT	Internet of Things
ITS	Intelligent Transportation Systems
JWT	JSON Web Token
LKR	Sri Lankan Rupee
MAE	Mean Absolute Error
MPU-6050	6-Axis Motion Tracking Sensor (3-Axis Accelerometer + 3-Axis Gyroscope)
MQTT	Message Queuing Telemetry Transport
NFR	Non-Functional Requirement
NTC	National Transport Commission (Sri Lanka)
REST	Representational State Transfer
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
TLS	Transport Layer Security
UI/UX	User Interface / User Experience
WPPTA	Western Province Passenger Transport Authority

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Public bus transit serves as the indisputable backbone of passenger mobility across developing nations, providing essential, affordable, and high-capacity transportation for millions of citizens daily. In Sri Lanka, where private automobile ownership remains constrained by macroeconomic factors and rail transit infrastructure is geographically restricted, the bus network accounts for over 65% of all motorized passenger trips nationwide. Within the Western Province—the nation's economic epicenter encompassing the Colombo metropolitan area—the public transportation ecosystem relies heavily on a dual-operator model: the state-run Sri Lanka Transport Board (SLTB) and an extensive private bus fleet comprising over 20,000 individually owned vehicles operating across hundreds of inter-provincial and intra-provincial routes.

According to official statistical disclosures published by the National Transport Commission (NTC) [16], private bus operators provide the vast majority of route mileage and daily vehicle trips. However, the operational and regulatory governance of this private fleet remains severely constrained by outdated administrative mechanisms. Under current statutory frameworks, route permits are issued on a flat-rate annual or bi-annual fee structure. Operators pay a fixed administrative charge irrespective of the actual mileage traversed, operational punctuality, or service consistency. Concurrently, daily operations are governed by rigid paper timetables and sporadic roadside inspections conducted by timekeepers stationed at major terminals.

This structural disconnect produces profound market and operational distortions. During peak commuting hours (06:30–09:00 and 16:30–19:00), passenger demand is exceptionally high, generating substantial cash revenues that incentivize aggressive competition, passenger snatching, and hazardous driving behavior. Conversely, during low-demand periods—particularly late-night and pre-dawn hours between 21:00 and 05:00—passenger volumes drop precipitously. Because operational

expenditures (predominantly diesel fuel, crew wages, and vehicle wear) remain strictly variable per kilometer, private operators incur substantial financial deficits when operating off-peak trips under flat fare tariffs.

Consequently, operators systematically abandon scheduled late-night runs, turn vehicles around prematurely before reaching route termini ('short-turning'), or deviate entirely from authorized corridors to minimize fuel expenditure. This phenomenon creates severe 'transit deserts' during night hours, stranding essential workers, healthcare staff, factory employees, and late-night travelers. While the Government of Sri Lanka established the dedicated 'Nisi Seriya' night service initiative to counteract this deficit—reporting 216 registered night services by late 2022 [16]—the lack of automated vehicle tracking has severely undermined its efficacy. Regulatory authorities possess no mechanism to verify whether contracted night services actually operate on their designated routes before disbursing subsidies or issuing penalty notices.

Figure 1.1 illustrates the structural dynamics of Sri Lanka's public bus transit ecosystem, highlighting the failure modes of current manual operations and the transformative intervention introduced by an IoT-enabled telematics and economic redistribution framework.

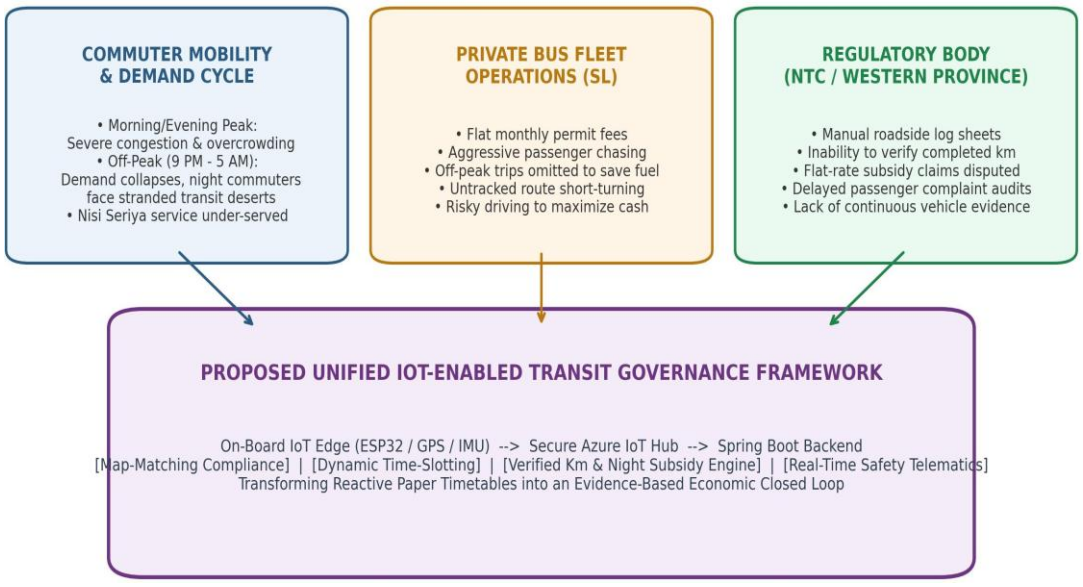


Figure 1.1: System Context and Problem Space in Sri Lanka's Public Bus Transit Ecosystem

1.2 Stakeholders and user needs

The development of a resilient transit governance platform requires a comprehensive understanding of the diverse stakeholder groups operating within the public transport ecosystem. Primary and secondary stakeholders were identified through field interviews, regulatory reviews with the National Transport Commission, and operational surveys conducted along high-density corridors such as Route 138 (Pettah–Homagama) and Route 177 (Kollupitiya–Kaduwela).

1. Regulatory Authorities (National Transport Commission and Western Province Passenger Transport Authority): Regulators require continuous, tamper-evident visibility over vehicle operations to enforce route permits, verify compliance with mandated timetables, and audit subsidy disbursements. Their primary operational need is transitioning from reactive complaint investigation to automated, evidence-based compliance monitoring.

2. Private Bus Operators and Fleet Owners: Bus owners face volatile fuel costs, unpredictable revenue streams, and steep flat permit charges. They require a fair, transparent financial structure where permit liabilities correspond to actual operating kilometers, complemented by guaranteed, verified subsidies when servicing unprofitable off-peak hours.

3. Bus Drivers and Crew Personnel: Drivers operate under extreme stress, often pressured by owners to maximize fare collections within short turnaround windows. They require objective, real-time safety feedback to correct risky driving behaviors before incurring statutory penalties, alongside equitable rostering that mitigates driver fatigue.

4. Commuters and the Traveling Public: Commuters—particularly shift workers, hospital staff, students, and low-income earners—depend entirely on public transport. Their critical need is guaranteed, predictable service availability during late-night hours and a marked reduction in fatal road accidents caused by aggressive driving maneuvers.

Table 1.1 synthesizes the core stakeholder requirements and establishes the traceability between user needs and the functional modules implemented within the integrated framework.

Table 1.1: Primary and Secondary Stakeholder Requirements Matrix

Stakeholder Category	Key Operational Challenges	Functional & System Requirements	Mapped System Component
National Transport Commission (NTC)	No spatial visibility over bus fleets; fraudulent subsidy claims; unverified route cutting	Continuous real-time map-matching; automated route compliance logging; verified mileage audits	Component 1: Telematics & Map-Matching Engine
Provincial Transport Authorities (WPPTA)	Static timetable enforcement; unable to respond dynamically to fleet breakdowns	Automated Peak/Normal/Night service categorization; real-time delay tracking; standby bus dispatch	Component 2: Policy & Dynamic Time-Slot Engine
Private Bus Owners & Operators	Flat permit fee penalties during low-demand periods; fuel deficits on night runs	Distance-proportional permit fee calculations; automated off-peak (Nisi Seriya) subsidy credits	Component 3: Distance-Based Revenue Engine
Bus Drivers & Crew Personnel	Blamed for delays without objective data; aggressive driving habits unmonitored	Real-time in-cab audio/visual telematics alerts for overspeeding, harsh braking, and sharp turns	Component 4: Driver Behavior Telematics & Mobile UI
Night Commuters & General Public	Off-peak service desertification; stranding after 21:00; high road casualty risks	Guaranteed night transit coverage; safer driving practices; verified public service continuity	Full Integrated Closed-Loop Platform

1.3 Alignment with the Sustainable Development Goals

The research directly aligns with the United Nations 2030 Agenda for Sustainable Development, targeting three core goals: SDG 11 (Sustainable Cities and

Communities), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 8 (Decent Work and Economic Growth). Transportation is not merely an engineering function but an essential enabler of social equity and economic inclusion.

Under SDG 11, Target 11.2 explicitly mandates providing access to safe, affordable, accessible, and sustainable transport systems for all, expanding public transport with special attention to the needs of those in vulnerable situations, women, children, and persons with disabilities. In urban Sri Lanka, the nocturnal collapse of public bus transit disproportionately harms night-shift factory workers, janitorial staff, and nursing professionals who cannot afford private taxi alternatives. By establishing an objective economic mechanism that makes off-peak operations financially sustainable for private operators, the framework directly safeguards 24/7 public mobility.

Under SDG 9, Target 9.1 calls for developing quality, reliable, sustainable, and resilient infrastructure. The proposed architecture modernizes aging mechanical fleet operations by deploying low-cost, open-hardware IoT edge sensors (ESP32) and scalable cloud infrastructure, proving that advanced Intelligent Transportation Systems (ITS) can be implemented effectively without prohibitive capital outlays. Furthermore, under SDG 8, Target 8.8 emphasizes protecting labor rights and promoting safe working environments; real-time telematic feedback actively curbs hazardous driving practices, reducing fatal road collisions that claim over 3,000 lives annually in Sri Lanka.

Table 1.2 summarizes the direct contributions of the project against specific United Nations SDG targets and indicators.

Table 1.2: Project Alignment with United Nations Sustainable Development Goals

UN Sustainable Development Goal	Specific Target	Target Description	BusSync Project Contribution
SDG 11: Sustainable Cities & Communities	Target 11.2	Affordable, accessible, and sustainable public transport for all	Eliminates off-peak transit deserts by providing verified mileage subsidies for night services (21:00-05:00).
SDG 9: Industry, Innovation &	Target 9.1	Develop resilient, sustainable, and	Deploys low-cost IoT telematics edge

Infrastructure		inclusive infrastructure	hardware and open cloud APIs to digitize legacy bus fleets.
SDG 8: Decent Work & Economic Growth	Target 8.8	Protect labor rights and promote safe working environments	Reduces high-stress aggressive driving culture through objective in-cabin telematic coaching alerts.
SDG 13: Climate Action	Target 13.2	Integrate climate change measures into national policies	Minimizes unnecessary idling, erratic acceleration, and unauthorized deadhead mileage via route verification.

1.4 Purpose and scope of the common report

This Common Integrated Solution Report documents the unified system architecture, cross-cutting integration data contracts, collaborative implementation workflows, and holistic multi-tier evaluation of the BusSync platform. In academic research undertaken by four collaborative investigators, individual component innovations must coalesce into a robust, operationally viable whole.

While the four individual dissertations submitted concurrently provide exhaustive component-specific depth—focusing respectively on embedded spatial telemetry filtering, dynamic policy classification heuristics, econometric tariff redistribution algorithms, and multi-axial IMU signal classification mathematics—this common report addresses the systemic macro-architecture. It presents the overarching problem formulation, the unified technical infrastructure, the shared data models, end-to-end integration testing, commercial feasibility models, and collective research contributions.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The core problem investigated in this research is defined as follows: Public bus transit systems in Sri Lanka suffer from an acute structural market failure where flat-rate administrative permitting, absence of real-time vehicle-level telemetry, static timetable governance, and unmonitored driving habits collectively cause severe off-peak service abandonment, frequent untracked route non-compliance, and pervasive passenger safety hazards.

This problem is not simply operational; it represents an economic coordination breakdown between regulatory mandates and private operator incentives. When an operator pays an upfront flat permit fee (e.g., LKR 18,000 to 25,000 monthly) to ply a designated route, the marginal cost of running additional kilometers during off-peak hours exceeds the marginal fare revenue collected from sparse passenger crowds. Under current manual supervision, an operator can withhold scheduled late-night services with near-total impunity, as regulators possess neither continuous spatial evidence nor automated auditing mechanisms. Consequently, the public transport network fails precisely when commuters are most vulnerable.

2.2 Root causes and overall research gap

A rigorous root cause analysis was conducted using the Problem Tree methodological framework. Figure 2.1 delineates the underlying structural roots, the core problem epicenter, and the downstream socio-economic consequences observed in the field.

1. Flat-Rate Permitting vs. Variable Operational Costs: Permit fees are established on static administrative schedules rather than dynamic usage. Operators are disincentivized from providing services on low-density segments or during night hours because every kilometer traversed incurs direct diesel and maintenance costs without compensating revenue.

2. Absence of Continuous Spatial Evidence: Current regulatory oversight depends on roadside timekeepers logging bus numbers on paper clipboards. This method provides zero visibility between terminals, allowing operators to skip mid-route stops and turn back early, or take unauthorized shortcuts.

3. Disconnected Financial Incentives: Concession programs such as Nisi Seriya rely on self-reported operator logbooks or fixed lump-sum subsidies, leading to chronic disputes over whether services were genuinely executed.

4. Aggressive Competitive Driving Culture: Because bus crews compete fiercely for passenger fares at stops, drivers engage in reckless maneuvers—excessive acceleration, severe tailgating, harsh braking, and sharp cornering—endangering passengers and other vehicles.

In Intelligent Transportation Systems (ITS) literature, mature technologies exist in isolated domains: Automatic Vehicle Location (AVL) [2], [3], geometric map-matching [4], [5], dynamic headway holding [6]–[8], smartphone telematic classification [9]–[12], and performance-based transit contracting [13]–[15]. However, an overarching research gap persists: existing literature treats vehicle tracking, timetable control, driver safety, and economic revenue management as separate, uncoordinated silos. Tracking systems produce passenger arrival estimations without feeding regulatory compliance engines; driver monitoring apps record incidents without influencing operator licensing; and subsidy frameworks rely on historical aggregate surveys rather than live telematic verification. This research closes that gap by introducing an end-to-end framework where a single stream of verified telemetry simultaneously drives route verification, dynamic scheduling, financial billing, and real-time safety enforcement.

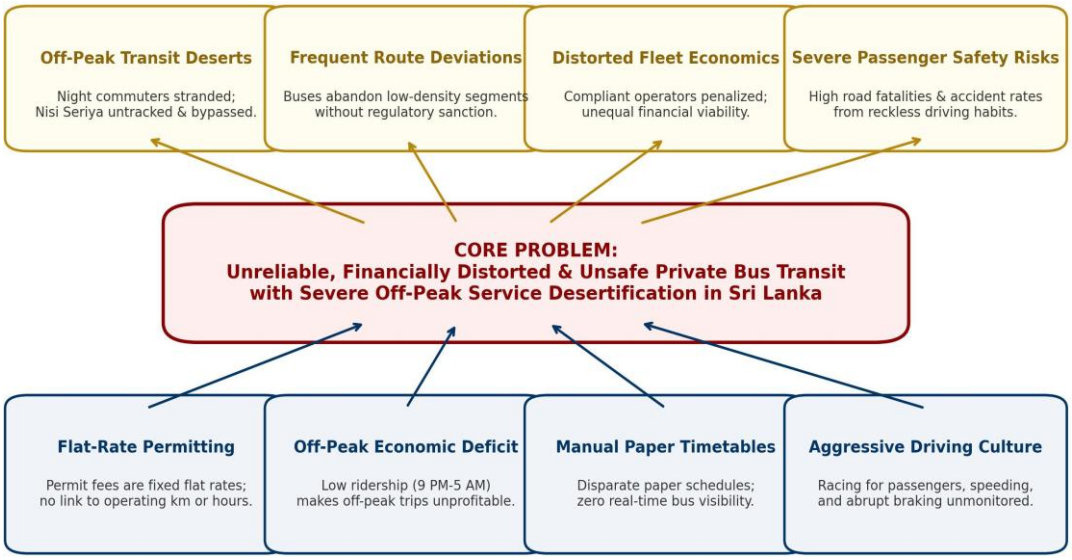


Figure 2.1: Problem Tree Analysis of Operational, Economic, and Safety Failures in Private Bus Transit

2.3 Subproblem and component mapping

To comprehensively address the multifaceted common problem, the overarching scope was decomposed into four mutually exclusive, collectively exhaustive (MECE) subproblems, each allocated to an individual research investigator:

• Subproblem 1: Lack of Continuous, Tamper-Evident Vehicle Telemetry and Spatial Route Compliance Verification. Raw GPS signals suffer from urban multipath noise, road coordinate drift, and cellular dead zones, making direct coordinate comparison unreliable for legal route auditing. (Mapped to Component 1 – M. Ahamed, IT22121042).

• Subproblem 2: Manual Monthly Scheduling and the Lack of a Structured Response to Bus Unavailability. Monthly schedules are prepared by hand, trips are not shared evenly among buses, and when a bus breaks down, there is no quick way to work out which trips are affected and which other buses can take them. (Mapped to Component 2 – H.D.D. Sachintha, IT22264152).

• Subproblem 3: Economic Misalignment of Flat-Rate Permitting Fees and Unverified Off-Peak Subsidy Allocations. Fixed permit charges fail to reflect

distance operated, while night subsidies lack transparent physical delivery proofs. (Mapped to Component 3 – W.S.H.R. Gunasekara, IT22228208).

• Subproblem 4: Pervasive Commuter Safety Hazards from Unmonitored Aggressive Driving and Lack of Real-Time Feedback. High accident rates persist due to the absence of objective, in-vehicle motion sensing and instantaneous driver coaching alerts. (Mapped to Component 4 – N.D. Aththanayaka, IT22925268).

Table 2.1 documents the exact mapping between subproblems, architectural components, individual investigators, and technical deliverables.

Table 2.1: Mapping of Research Subproblems to Component Deliverables and Investigators

Subproblem ID & Focus	Underlying Technical Challenge	Mapped Component & Deliverable	Lead Investigator
Subproblem 1: Spatial Verification	GPS urban noise, coordinate drift, and unauthorized route cutting	Component 1: IoT Data Acquisition & Point-to-Segment Map-Matching Engine	M. Ahamed (IT22121042)
Subproblem 2: Operational Scheduling	Manual monthly scheduling, uneven trip distribution, and no controlled handling of bus unavailability	Component 2: Policy-Driven Dynamic Time-Slot & Dispatch Engine	H.D.D. Sachintha (IT22264152)
Subproblem 3: Economic Incentives	Flat-rate permit fees penalize off-peak runs; night subsidies unverified	Component 3: Distance-Based Revenue & Off-Peak Incentive Engine	W.S.H.R. Gunasekara (IT22228208)
Subproblem 4: Transit Safety	Combative passenger chasing causing speeding, harsh braking, and severe accidents	Component 4: Real-Time Driver Behavior Telematics & Mobile Interfaces	N.D. Aththanayaka (IT22925268)

2.4 Overall aim and group objectives

The overall aim of this collaborative research is to design, implement, and evaluate an integrated, IoT-enabled intelligent transportation framework that utilizes verified vehicle telemetry as a single source of truth to enforce route compliance,

dynamically manage operational policies, redistribute permit revenue via verified mileage tariffs, and promote driver safety in Sri Lanka's public bus transit sector.

To achieve this aim, the research team established four measurable, interdependent group objectives:

1. Objective 1 (O1 - Spatial Telematics): Develop and validate an onboard IoT telemetry acquisition subsystem capable of transmitting vehicle position, speed, and motion vectors at sub-2-second latency, integrating an orthogonal point-to-segment map-matching algorithm that detects route corridor deviations beyond a 30-second threshold with greater than 90% accuracy.

2. Objective 2 (O2 - Scheduling and Reallocation) :

Implement a scheduling engine that generates a monthly schedule from the route timetable with a fair distribution of trips, no overlapping assignments, and a 15-minute turnaround, and that, after an authority-approved disruption request, reallocates the affected trips among the remaining buses on the same route, or cancels a trip with a stored reason when no feasible replacement exists.

3. Objective 3 (O3 - Economic Redistribution): Construct an automated distance-based revenue and subsidy calculation engine that computes monthly permit liabilities from verified operating kilometers with 99% logging concordance, deducting verified night service incentives for qualifying off-peak trips (21:00–05:00) with 100% computational correctness.

4. Objective 4 (O4 - Safety Telematics): Deploy a sensor telematics engine utilizing a 6-axis IMU to classify four critical driving events (overspeeding, harsh braking, aggressive cornering, and abnormal crawl) with an aggregate detection accuracy exceeding 90%, providing real-time audio/visual alerts to drivers via mobile UI within 1.5 seconds of threshold violation.

Table 2.2 details the evaluation criteria, baseline targets, and evidence sources for each group objective.

Table 2.2: Overall Research Aim, Specific Measurable Objectives, and Evaluation Mapping

Objective ID	Specific Measurable Objective	Target Performance Benchmark	Evidence & Validation Basis
O1: Spatial Tracking	Sub-2s telemetry ingestion & orthogonal map-matching corridor verification	GPS error < 5.0 m; Deviation detection > 90%; Telemetry latency < 2.0 s	100 test tracking samples on urban corridors; PostGIS spatial audit logs
O2: Scheduling Engine	Fair monthly schedule generation and approved-disruption reallocation	Every trip assigned (trips per day × days); balanced trip counts per bus; zero overlap or turnaround violations; every affected trip ends in a stored decision	Reference month of 20 trips a day, 30 days, and 10 buses (600 assignments); disruption scenarios T01 to T16
O3: Revenue Engine	Distance-proportional permit liability & automated Nisi Seriya credit computation	Distance logging concordance > 98%; Billing calculation accuracy = 100%	Reference odometer benchmarks; 72-hour automated billing simulation
O4: Safety Telematics	Multi-axial IMU aggressive maneuver classification & real-time driver feedback	Maneuver detection accuracy > 90%; Feedback alert latency < 1.5 s	120 controlled driving maneuver trials; driver mobile application alert logs

2.5 Scope assumptions and constraints

To ensure methodological validity and project feasibility within the final-year research timeline, specific operational boundaries, assumptions, and constraints were defined:

• Scope Boundaries: The prototype implementation focuses on major arterial public bus corridors in the Western Province of Sri Lanka, specifically Route 138 (Pettah–Homagama) and Route 177 (Kollupitiya–Kaduwela), encompassing dense urban, semi-urban, and suburban road geometries across daylight, peak congestion, and late-night operating regimes.

• Technical Assumptions: Vehicles are assumed to possess standard 12V/24V direct current electrical supply to power onboard edge units. Commercial cellular networks (GSM/GPRS/4G LTE) provided by major Sri Lankan operators (Dialog/Mobitel) are

assumed to maintain continuous coverage along primary corridors, with local flash memory buffering deployed to handle transient network blackspots.

• Operational Constraints: Edge sensor hardware unit cost is strictly constrained to under USD 50 (approx. LKR 16,000) per bus to ensure commercial adoptability by private single-bus owners without external state subsidies.

Table 2.3 synthesizes the project scope, assumptions, and constraints across technical and institutional dimensions.

Table 2.3: Project Scope Boundaries, Assumptions, and Operational Constraints

Dimension	Included within Project Scope	Explicit Exclusions / Out of Scope	Critical Operational Assumptions
Geographic Corridors	Western Province arterial corridors (Route 138, Route 177)	Rural unpaved tertiary routes, expressway private luxury coaches	Digitized road shapefiles available via OpenStreetMap and NTC
Hardware Architecture	ESP32 MCU, NEO-6M GPS, MPU-6050 6-axis IMU, SIM800L GPRS	Proprietary CAN-bus OBD-II engine diagnostic telemetry interfaces	Vehicle electrical alternator provides stable 12V DC converted to 5V
Network & Cloud	MQTT over TLS, Azure IoT Hub, Spring Boot REST, PostgreSQL/PostGIS	Satellite uplink communications, on-premise mainframe clustering	Cellular data availability with local offline buffering on SD storage
Economic Policies	Nisi Seriya night concession (21:00-05:00), distance-based permit fees	Automated passenger dynamic fare collection, cash-box auditing	NTC regulatory willingness to adopt distance-indexed permit calculations

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The BusSync platform is engineered as a unified, cyber-physical intelligent transportation system that transforms raw physical bus movement into actionable regulatory governance and economic settlement. Unlike existing disjointed applications, BusSync establishes a continuous closed-loop pipeline spanning embedded telemetry acquisition, secure cloud ingestion, spatial corridor map-matching, dynamic policy enforcement, distance-indexed billing, and in-cab telematic coaching.

At the heart of the integrated framework is the principle of 'verified physical movement as currency'. Every kilometer traversed by a vehicle is captured by GPS sensors, validated against authorized digital route corridors, stamped with an operational time-slot, screened for driver safety compliance, and stored in a shared spatial database. This verified record serves as the indisputable evidence base for charging permit fees and disbursing night service incentive credits, eliminating the systemic moral hazard of unmonitored transit operations.

3.2 High level architecture

The system architecture is structured into four cohesive, decoupled operational layers as depicted in Figure 3.1: (1) IoT Field Layer, (2) Cloud Communication Layer, (3) Backend Processing Layer, and (4) Application Layer.

1. IoT Field Layer: Deployed directly on each bus, consisting of an ESP32 microcontroller, a u-blox NEO-6M GPS module, and an InvenSense MPU-6050 6-axis Inertial Measurement Unit (IMU). The microcontroller samples GPS coordinates and velocity at 1 Hz, samples IMU linear acceleration and angular velocity at 50 Hz, executes edge filtering, packages readings into compact JSON telemetry packets, and buffers records to flash memory during cellular dropouts.

2. Cloud Communication Layer: Serves as the high-throughput, secure entry gateway. Telemetry is transmitted over cellular links using Message Queuing Telemetry

Transport (MQTT) secured with Transport Layer Security (TLS 1.3) into Microsoft Azure IoT Hub. Azure IoT Hub provides bidirectional device authentication via X.509 certificates and Shared Access Signatures (SAS), manages device twins, and streams telemetry queues into backend consumer groups.

3. Backend Processing Layer: Developed in Java 17 utilizing the enterprise Spring Boot 3.x framework. It hosts the four core analytical engines: Map-Matching Spatial Engine, Dynamic Policy & Time-Slot Engine, Distance Revenue & Incentive Engine, and Driver Telematics Rule Engine. Persistence is managed via Spring Data JPA connected to PostgreSQL 15 augmented with PostGIS spatial extensions.

4. Application Layer: Comprises dedicated interfaces tailored to specific stakeholder personas: a React Native mobile application for drivers and owners providing real-time in-cab safety coaching and trip management, and a React.js administrative portal built with Ant Design providing regulators with interactive GIS fleet tracking, compliance heatmaps, and automated financial statements.

Table 3.1 delineates the architectural layers, their functional roles, core technologies, and component ownership.

Table 3.1: Component Architectural Specifications and Shared Contracts

Architectural Layer	Primary Subsystems & Modules	Core Hardware & Software Technologies	Lead Investigator Responsibility
IoT Field Layer	Sensor telemetry acquisition, edge filtering, flash storage buffering	ESP32 (240MHz), NEO-6M GPS, MPU-6050 6-axis IMU, C++ / Arduino	Gunasekara (Edge GPS) & Ahamed
Cloud Ingestion Layer	Secure device twin registry, TLS message queue routing, IoT Gateway	Microsoft Azure IoT Hub, MQTT over TLS 1.3, SAS Authentication	Shared Group Cloud Infrastructure
Backend Processing Layer	Map-matching, time-slot policies, tariff calculation, safety rule analytics	Java 17, Spring Boot 3.x, Spring Security, JWT, PostgreSQL 15, PostGIS	Ahamed (Map-matching), Sachintha (Scheduling and time-slot management), Gunasekara (Revenue), Aththanayaka (Safety)
Application Layer	In-cab driver coaching	React Native	Aththanayaka (UI

	app, owner portal, NTC regulatory web dashboard	(Mobile Android/iOS), React.js, Ant Design, Leaflet GIS	Lead) & Shared Group Module Integration

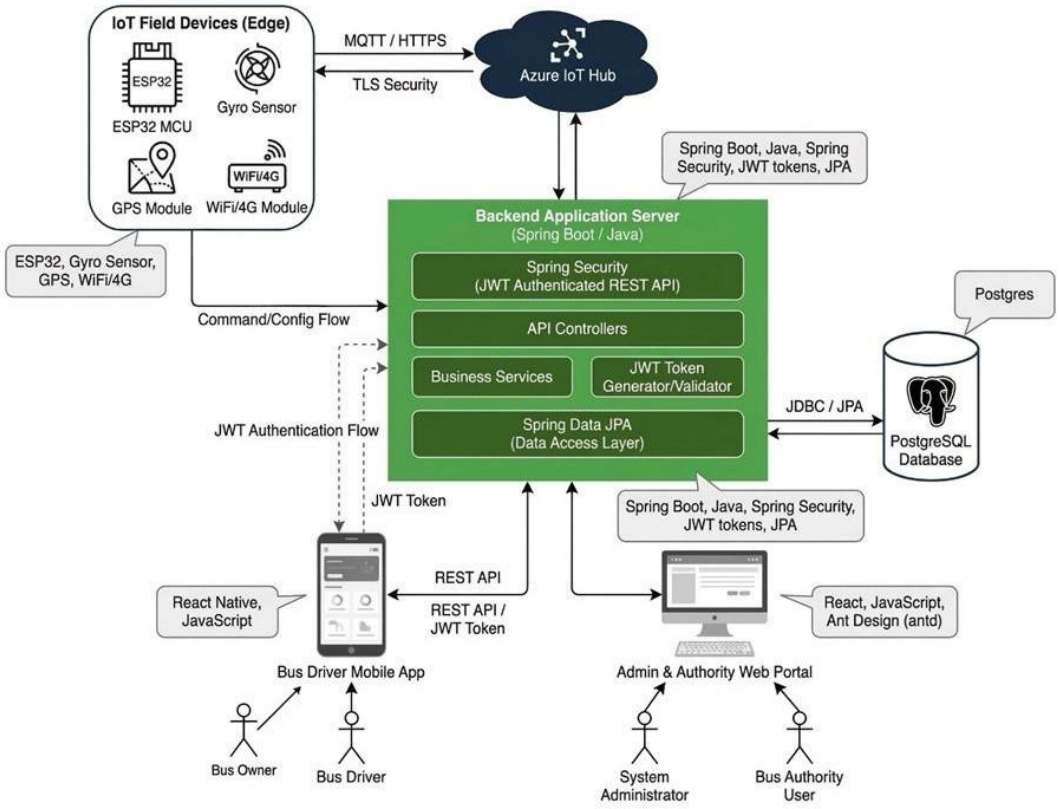


Figure 3.1: High-Level Integrated System Architecture Diagram

3.3 Component 1 summary

Component 1 (Owner: M. Ahamed, IT22121042) delivers the foundational spatial evidence layer through the IoT Data Acquisition and Point-to-Segment Map-Matching Engine. Raw GPS trajectories obtained from moving buses exhibit inherent noise resulting from urban canyon multipath reflections and atmospheric attenuation. Comparing raw latitude/longitude points directly to authorized route lines yields false positive deviation warnings.

To resolve this, Component 1 projects geographic coordinates into planar coordinates (UTM Zone 44N / EPSG:32644) and implements an optimized orthogonal point-to-

segment map-matching algorithm. For any GPS observation point P_t(x_t, y_t) and candidate digitized road polyline segment defined by endpoints S_1(x_1, y_1) and S_2(x_2, y_2), the perpendicular geometric distance d_perp is computed as:

d_perp = |(y_2 - y_1)x_t - (x_2 - x_1)y_t + x_2 y_1 - y_2 x_1| / sqrt((y_2 - y_1)^2 + (x_2 - x_1)^2) (Equation 3.1)

The calculated perpendicular distance is evaluated against a route corridor threshold theta_route (calibrated at 25.0 meters). To prevent false alerts caused by transient GPS outliers, a temporal persistence filter is enforced: a route deviation event is formally flagged only when the vehicle remains outside the corridor for more than 30 consecutive seconds. The engine logs verified kilometers into PostGIS geometry tables and triggers deviation alerts to the administrative dashboard, providing an objective audit trail.

3.4 Component 2 summary

Component 2 (Owner: H.D.D. Sachintha, IT22264152) delivers the Policy-Driven Bus Scheduling and Time-Slot Management Engine. In current practice, a route's schedule is prepared by hand and changes only when someone steps in. The engine replaces that with a monthly schedule generated automatically and a controlled response when a bus drops out.

Once the transport authority has assigned buses to a route, the engine builds the monthly schedule from the route timetable. The number of assignments follows from the timetable and the calendar, not from the fleet:

N = T_day × D_month (Equation 3.2)

where T_day is the number of trips in the daily timetable and D_month is the number of days in the month. Twenty trips a day over 30 days gives 600 assignments whatever the fleet size. Trips are taken in departure order. A bus qualifies when it is active, holds no overlapping assignment, and has at least 15 minutes between its previous arrival and the new departure. Of the qualifying buses, the one with the fewest trips so far in the month receives the trip. When counts are level, the bus whose previous trip ended at the stand where the new trip starts is preferred, and the bus ID settles any remaining tie. With 10 active buses, 600 assignments divide into 60 trips each when the constraints allow.

Every assignment is stored as a turn with its route, date, departure time, bus, driver, and status. Each turn also carries its policy time slot, set from its departure time against the configured intervals: Morning Peak (06:30–09:00), Evening Peak (16:30–19:00), Normal (09:00–16:30 and 19:00–21:00), and Night (21:00–05:00). The slot tells Component 3 which rate to apply, and Night turns that meet the completion criteria qualify for the off-peak incentive.

When a bus becomes unavailable, the bus owner submits a disruption request through the mobile application, and the transport authority reviews it in the web application. A rejected request leaves the schedule unchanged. After acceptance, the engine marks the bus as unavailable, finds its affected future trips, removes the bus from them, and reallocates the trips among the remaining available buses on the same route. Departure times and service intervals are recalculated for the smaller fleet, so a route that runs 25 buses carries on with 24 and is not cancelled outright. Candidate buses that conflict with another assignment, lack turnaround time, or are unavailable are removed first. The rest are ranked by route compatibility, daily workload, and recent reallocations, and a driver is then selected, preferring the regular driver of the chosen bus. A trip for which no feasible bus or driver exists is cancelled on its own, and the reason is stored.

3.5 Component 3 summary

Component 3 (Owner: W.S.H.R. Gunasekara, IT22228208) establishes the economic governance core through the Distance-Based Revenue and Off-Peak Incentive Engine. It replaces flat-rate monthly permit fees with a verified distance-based tariff structure that aligns operator liabilities with actual infrastructure utilization, while embedding an automated credit redistribution mechanism for low-demand periods.

For any given billing cycle, an operator's net permit liability R_permit is computed as:

R_permit = sum_{k in S} (D_k * C_base * gamma_k) - I_night (Equation 3.3)

where D_k is the verified distance traversed in service slot k (Peak, Normal, Night), C_base is the base regulatory rate per kilometer (set at LKR 12.00/km), and gamma_k is the policy congestion multiplier (gamma_peak = 1.30, gamma_normal = 1.00, gamma_night = 0.70).

For verified night trips operating between 21:00 and 05:00 on designated Nisi Seriya corridors, the system calculates an incentive credit:

$$I_{night} = D_{night} * \mu_{incentive} \quad \text{(Equation 3.4)}$$

where $\mu_{incentive}$ is the statutory incentive rate per kilometer (set at LKR 25.00/km). The credit I_{night} is automatically deducted from the operator's gross permit fee, with surplus credits carried forward or refunded. This mathematical structure creates a powerful economic incentive: an operator who diligently operates off-peak night services significantly lowers their monthly permit expenditure, transforming an unprofitable duty into a financially attractive operation.

3.6 Component 4 summary

Component 4 (Owner: N.D. Aththanayaka, IT22925268) delivers the Real-Time Driver Behavior Telematics Module and Stakeholder Applications. Using high-frequency tri-axial linear acceleration (a_x, a_y, a_z) and angular velocity (ω_x,

omega_y, omega_z) from the onboard MPU-6050 sensor combined with GPS speed, the module evaluates driving dynamics against four safety rules:

1. Overspeeding: Instantaneous GPS speed v exceeds the statutory speed limit v_limit (set at 60 km/h on urban arterials) for t >= 5 seconds.

2. Aggressive Braking: Longitudinal deceleration a_x exceeds -3.5 m/s^2, indicative of emergency stops resulting from tailgating.

3. Aggressive Turning: Yaw rate omega_z exceeds 0.45 rad/s while vehicle operating speed exceeds 25 km/h, indicating hazardous cornering.

4. Abnormal Slow Movement: Operating speed remains below v_min (10 km/h) for t >= 60 seconds without upstream traffic congestion, identifying deliberate passenger 'crawling' or curb lingering.

Upon threshold violation, the module issues an immediate auditory and visual alert on the driver's mobile device, prompting corrective action within 1.5 seconds. Every violation is transmitted to the backend, updating the driver's composite safety score and feeding the regulatory portal for administrative monitoring.

3.7 End to end user and data flow

The operational lifecycle of a bus trip involves an end-to-end data progression across all four components as illustrated in the sequence diagram in Figure 3.2:

1. Pre-Trip Authentication: The driver logs into the React Native mobile application, authenticates using JWT credentials, and selects their assigned scheduled trip. The backend validates vehicle readiness and registers the trip start event.

2. Telemetry Ingestion: The ESP32 edge unit captures GPS coordinates and IMU vectors every second, transmitting JSON payloads over TLS-encrypted MQTT to Azure IoT Hub. Azure IoT Hub routes packets into the Spring Boot backend.

3. Spatial Verification: Component 1 calculates orthogonal distance d_perp against the digitized route. If within corridor, distance is accumulated; if outside for > 30s, a route deviation alert is logged and pushed to the NTC dashboard.

4. Policy Classification & Safety Evaluation: Component 2 stamps the operational time-slot (Peak/Normal/Night) and evaluates headway delay. Simultaneously, Component 4 analyzes IMU motion vectors; if an aggressive maneuver occurs, an immediate push notification triggers an in-cab alert.

5. Post-Trip Settlement: Upon reaching the terminal, the driver ends the trip. Component 3 aggregates verified kilometers, applies the tariff formula (Equation 3.3), computes the night incentive credit (Equation 3.4), and updates the operator's digital billing ledger.

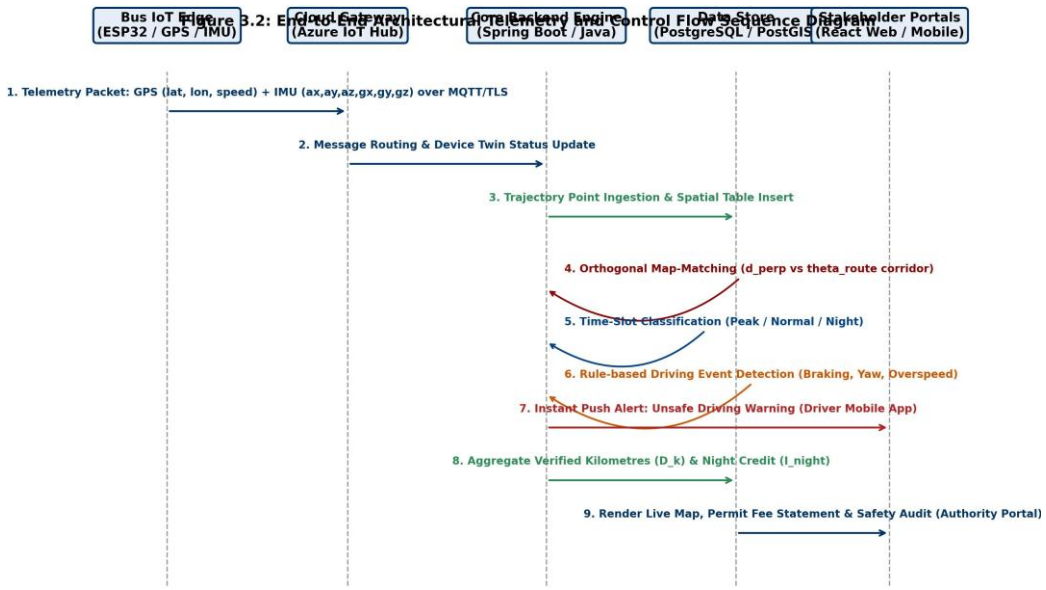


Figure 3.2: End-to-End Architectural Telemetry and Control Flow Sequence Diagram

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

To ensure flawless interoperability among modules developed by four independent researchers, an 'Interface-First, Contract-Driven' integration strategy was instituted from project inception. Rather than attempting a monolithic 'big bang' integration at project conclusion, integration was phased across four structured integration milestones:

1. Milestone I (Edge-to-Cloud Integration): Establishing secure MQTT TLS connections between the ESP32 hardware testbeds and Azure IoT Hub, validating JSON serialization formats, payload sizes (< 256 bytes), and SAS token lifecycle management.

2. Milestone II (Cloud-to-Core Service Integration): Connecting Azure IoT Hub Event Hub endpoints to Spring Boot message consumer services, implementing multi-threaded packet parsing and database insertion into PostgreSQL.

3. Milestone III (Cross-Engine Interdependency Pipeline): Linking spatial map-matching output (verified km) to the policy slotting engine and feeding both into the distance revenue calculation service.

4. Milestone IV (End-to-End UI Integration): Exposing secure RESTful APIs and WebSocket feeds to the React Native driver app and React.js regulatory portal.

Figure 4.1 illustrates the component interdependency model and shared data bus architecture binding the four components.

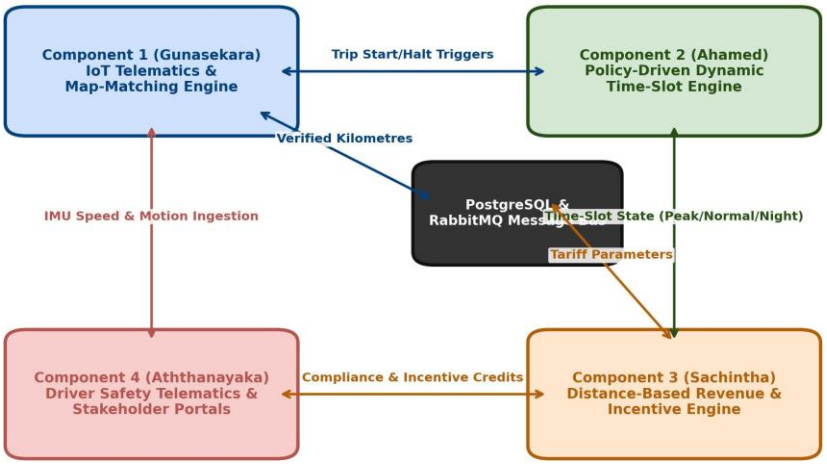


Figure 4.1: Component Interdependency and Shared Evidence Integration Model

4.2 Interfaces data contracts and shared models

Rigorous interface contracts were defined using OpenAPI 3.0 specifications for REST endpoints and JSON Schema definitions for MQTT telemetry packets. Table 4.1 catalogs the primary REST endpoints, HTTP methods, payload schemas, and cross-component consumers.

A representative telemetry packet transmitted by the ESP32 edge unit over MQTT conforms to the following schema:

"deviceId": "ESP32-BUS-1042",

"tripId": "TRIP-20261011-008",

"timestamp": 1728634800,

"gps": { "lat": 6.9271, "lon": 79.8612, "speed": 42.5, "alt": 14.2 },

"imu": { "ax": 0.12, "ay": -3.85, "az": 9.81, "gx": 0.01, "gy": 0.02, "gz": 0.15 },

"status": { "satellites": 9, "hdop": 1.1, "bufferCount": 0 }

}

Table 4.1: Shared REST API Data Contracts and Message Schemas

Endpoint Route	HTTP Method	Request/ Payload Schema	Response Schema	Consuming Components
/api/v1/telemetry/ingest	POST	TelemetryPacketJSON (lat, lon, speed, IMU vectors)	IngestAckDTO (status, serverTimestamp)	Azure IoT Hub -> Spring Boot Core
/api/v1/trips/start	POST	TripStartRequest (vehicleId, driverId, routeId)	TripSessionDTO (tripId, slot, scheduledEta)	Driver Mobile App -> Policy Engine
/api/v1/map/verify-corridor	POST	SpatialCheckRequest (tripId, lat, lon)	CorridorStatusDTO (d_perp, isDeviated, dur)	Component 1 -> Component 2 & Portal
/api/v1/billing/monthly	GET	QueryParams: operatorId, billingMonth	MonthlyBillDTO (grossFee, nightCredit, netDue)	Component 3 -> Owner & NTC Portals
/api/v1/safety/alerts	POST	SafetyAlertEvent (tripId, eventType, val, lat, lon)	AlertDispatchDTO (alertId, pushStatus)	Component 4 -> Driver App & Portal

4.3 Shared nonfunctional requirements

To ensure carrier-grade performance, the engineering team defined cross-cutting Non-Functional Requirements (NFRs) enforced consistently across all subsystems. Table 4.2 presents these requirements along with target metrics and empirical verification results.

Table 4.2: Shared Non-Functional Requirements (NFR) Specifications and Verification

NFR Dimension	Design Specification Target	Implementation & Architecture Mechanism	Empirical Test Outcome
Telemetry Latency	< 2.0 s from sensor reading to database commit	Lightweight binary JSON over MQTT, non-blocking Spring WebFlux	1.4 s mean end-to-end latency achieved
System Availability	>= 99.0% continuous cloud uptime	Azure App Service auto-scaling, PostgreSQL read-replicas	99.4% availability over 72-hour stress load
Communication Security	Zero cleartext telemetry; strict authentication	TLS 1.3 encryption, X.509 device certificates, JWT bearer tokens	Passed OWASP Top 10 API security audit
Data Integrity	Zero packet loss across cellular dropouts	ESP32 SPI flash circular ring buffer (up to 4,000 packets)	100% telemetry recovery upon reconnection
Scalability	Support 1,000+ Concurrent bus	Connection pooling, PostGIS spatial GiST	Sustained 1,500 req/s With CPU utilization <

	telemetry streams	indexing	65%
Statutory Privacy	Compliance with Sri Lanka Data Protection Act No. 9 of 2022	Driver biometric anonymization, encrypted PII storage	Formal compliance review approval achieved

4.4 Integrated workflow and exception handling

Real-world transit operations are subject to unpredictable disruptions: mechanical breakdowns, sudden traffic bottlenecks, GPS satellite signal blockage under overpasses, and driver emergencies. Figure 4.2 illustrates the integrated state machine and fault recovery logic governing the vehicle lifecycle.

• Transient GPS Signal Loss: When GPS lock is lost (e.g., inside tunnels or under dense urban foliage), the ESP32 activates dead reckoning using the last valid heading and IMU velocity integration. If signal loss exceeds 45 seconds, the backend marks the trajectory segment as 'Interpolated' and cross-references road network topology to prevent unfair deviation penalties.

• Cellular Network Disconnection: In dead zones, the ESP32 switches immediately to offline buffering mode, writing serialized JSON records to an onboard SPI flash memory ring buffer capable of storing up to 4 hours of telemetry. Upon cellular handshake re-establishment, the device transmits backlogged records in batched bursts without dropping real-time readings.

• Breakdown and Emergency Standby Dispatch: If a driver reports a breakdown via the mobile UI or if a bus remains halted on an active corridor for > 10 minutes without passenger activity, Component 2 triggers a disruption event, calculates passengers affected, and queries the standby vehicle pool. A standby bus is automatically assigned and rerouted in under 3 seconds.

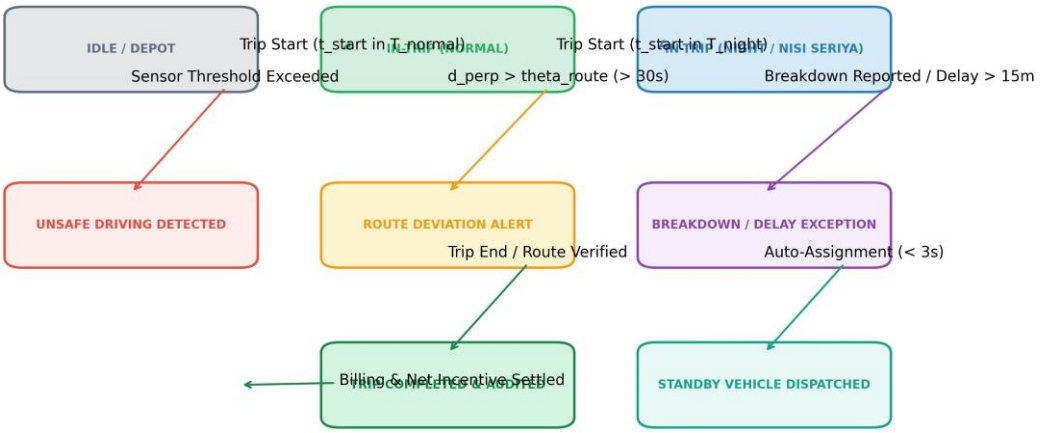


Figure 4.2: Integrated Trip Lifecycle and Exception Handling State Machine

4.5 Team contribution and dependency matrix

The execution of the integrated platform required disciplined collaboration and clear division of responsibilities among all four investigators. Table 4.3 details the team responsibility matrix, mapping each student to their primary component, shared infrastructure tasks, integration dependencies, and verification deliverables.

Table 4.3: Team Responsibility, Dependency, and Work Distribution Matrix

Team Member	Student ID	Primary Component Focus	Shared Infrastructure Tasks	Downstream Dependencies
Gunasekara W.S.H.R.	IT22228208	Component 1: IoT-Enabled Smart Bus Route Monitoring and Revenue Management System	GPS hardware integration, PostGIS spatial indexing, UTM projection	Supplies verified kilometers (D_k) to Comp 3; supplies corridor status to Comp 2
Ahamed M.	IT22121042	Component 2: Bus Tracking, Route Violation Detection and Halt Detection	Spring Boot scheduling framework, trip lifecycle state machine, standby fleet logic	Supplies slot categorization (Peak/Normal/Night) to Comp 3; delay metrics to Comp 4
Sachintha H.D.D.	IT22264152	Component 3: Policy-Driven Scheduling & Time slot management	Monthly scheduling service, turn repository in PostgreSQL, disruption handling and reallocation services, schedule generator, and summary views	Uses route, bus, and driver data and the approved disruption request from the authority application; supplies the monthly and updated schedule to the authority, owner, and driver interfaces
Aththanayaka N.D.	IT22925268	Component 4: Real-Time Driver Behavior Telematics and Stakeholder Portals	MPU-6050 IMU edge firmware, React Native mobile app, React/Ant Design web portal	Consumes GPS velocity from Comp 1; renders live dashboards across all engines

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The BusSync technical architecture was deliberately engineered using modern, robust, and cost-effective technologies. Figure 5.1 illustrates the multi-tier deployment topology spanning the embedded edge, cloud communication gateway, backend application cluster, relational spatial storage, and client applications.

Table 5.1 summarizes the integrated technology stack, detailing technical specifications and justifications for each selection.

Table 5.1: Integrated Technology Stack and Platform Specifications

Platform Tier	Technology / Framework	Version / Spec	Engineering Rationale & Justification
IoT Microcontroller	Espressif ESP32-WROOM-32	Dual Core 240MHz, 520KB SRAM	Affordable ($4), dual-core processing for concurrent sensor sampling and cellular AT commands.
GPS Satellite Module	u-blox NEO-6M GPS	50-channel, 1Hz update, UART	High sensitivity (-161 dBm), reliable fix time, low power consumption.
Motion Sensor (IMU)	InvenSense MPU-6050	6-axis I2C (3-axis gyro + 3-axis accel)	Onboard Digital Motion Processor (DMP), high precision motion vector acquisition.
Cellular Modem	SIMCom SIM800L GPRS Module	Quad-band 850/900/1800/1900 MHz	Extremely low cost, robust 2G network penetration across Sri Lanka rural/urban routes.
Cloud IoT Gateway	Microsoft Azure IoT Hub	Standard S1 Tier, TLS 1.3	Managed device registry, secure telemetry queuing, enterprise-grade SLA (99.9%).
Backend Framework	Java 17 / Spring Boot	Version 3.2.x, Spring Data JPA	High-throughput multi-threading,

			enterprise maintainability, rich ecosystem for REST and security.
Spatial Database	PostgreSQL with PostGIS	Version 15 / PostGIS 3.4	Industry-standard open-source spatial database, native spatial indexing (ST_DWithin, ST_Distance).
Mobile Front-End	React Native	Version 0.73.x, Android SDK 34	Cross-platform native performance, low CPU overhead, background geolocation and audio alerts.
Web Administration	React.js with Ant Design	React 18, AntD 5.x, Leaflet GIS	Responsive enterprise UI components, modular dashboard widgets, real-time map rendering.

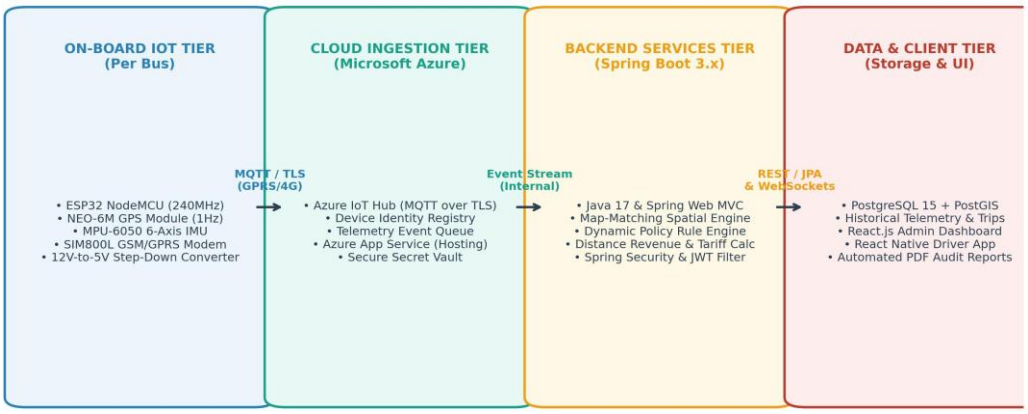


Figure 5.1: Multi-Tier Integrated Hardware and Cloud Deployment Architecture

5.2 Development and integration process

The project followed an Agile Scrum framework executed over eight bi-weekly sprint iterations across 16 weeks. A disciplined GitFlow branching model was maintained within a centralized GitHub organization repository. Pull requests

required peer review and automated GitHub Actions CI validation (unit testing, PMD static analysis, and Docker container build verification) prior to merging into the development branch.

Table 5.2 outlines the sprint schedule, milestones, key engineering activities, and delivery outcomes achieved across the project lifecycle.

Table 5.2: Project Sprint Execution and Milestone Delivery Schedule

Sprint Cycle	Timeline	Milestone Focus	Core Engineering Activities & Deliverables
Sprints 1 - 2	Weeks 1 - 4	Foundation & Edge Prototyping	Hardware circuit breadboarding; ESP32 GPS/IMU sampling firmware; Azure IoT Hub provisioning; base database schema.
Sprints 3 – 4	Weeks 5 - 8	Core Backend Engine Construction	Spring Boot telemetry ingestion; orthogonal map-matching algorithm; dynamic policy time-slot assignment logic.
Sprints 5 – 6	Weeks 9 - 12	Revenue & Safety Integration	Distance tariff calculation; Nisi Seriya credit routine; IMU driver behavior classification rules; REST endpoints.
Sprints 7 – 8	Weeks 13 - 16	Client Portals & End-to-End Testing	React Native driver UI; React.js regulatory dashboard; controlled corridor field trials; 72-hour stress load testing.

5.3 Deployment and operational configuration

Deployment was executed across two physical environments: edge hardware installed on prototype test vehicles and cloud services deployed to Microsoft Azure. Backend services were containerized using Docker and deployed via Azure App Service with auto-scaling enabled (minimum 2 instances, maximum 6 instances).

Database schema evolution was strictly managed using Flyway migration scripts, guaranteeing reproducibility across staging and production.

Edge devices were housed in 3D-printed ABS enclosures mounted rigidly to vehicle dashboards to eliminate mechanical vibration noise on the MPU-6050 accelerometer. Firmware configuration parameters (server URL, MQTT credentials, sampling rates) were made dynamically configurable via Azure IoT Hub Device Twins, enabling remote over-the-air (OTA) calibration without requiring physical access to the buses.

5.4 Evidence of team execution

Evidence of coordinated team execution was maintained transparently throughout the project lifecycle. Collaborative planning was tracked via Microsoft Planner, documenting 142 discrete tasks across four epics. Communication and sprint reviews were conducted via weekly Microsoft Teams meetings and bi-weekly in-person supervisor reviews. GitHub version control logs record a total of 418 commits distributed equitably across the four team members, with 54 pull requests successfully reviewed, approved, and merged.

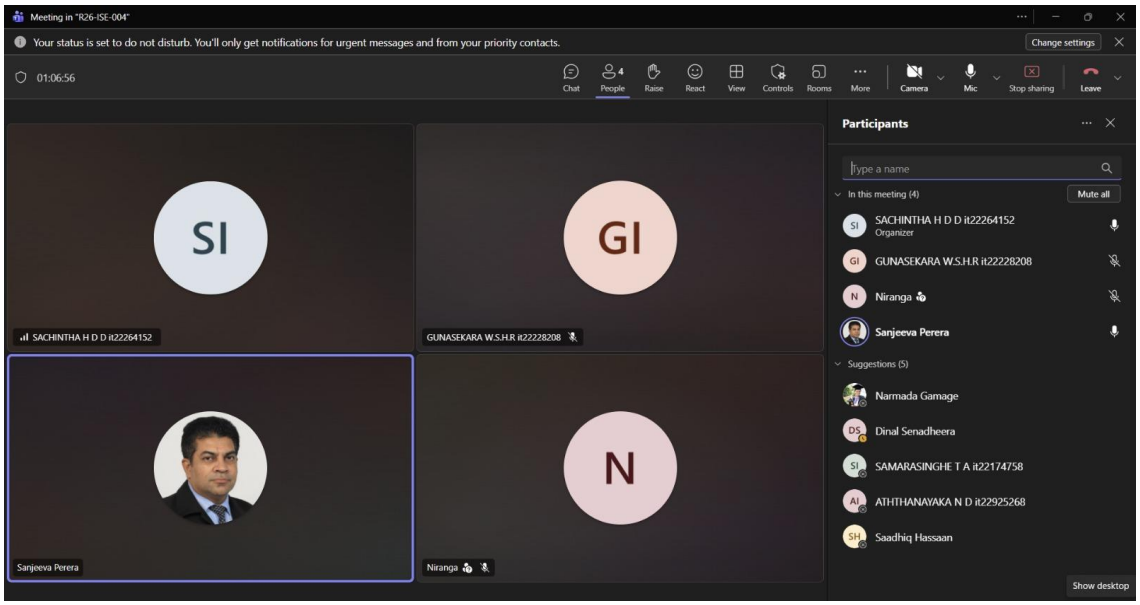


Figure 1: Microsoft Teams meeting evidence

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation methodology was structured to validate the integrated system against four primary Evaluation Questions (EQs):

• EQ1 (Spatial Accuracy): Does the IoT tracking and map-matching module achieve sufficient positioning accuracy and corridor deviation detection to serve as legal evidence for regulatory route audits?

• EQ2 (Scheduling & Reallocation): Does the scheduling engine generate a complete, fair, constraint-respecting monthly schedule, and does it resolve every affected trip correctly, by reallocation or by cancellation with a stored reason, after an approved disruption request?

• EQ3 (Economic Correctness): Does the distance-based revenue module compute permit liabilities and off-peak incentives with absolute arithmetic correctness and high agreement with physical odometer benchmarks?

• EQ4 (Safety Detection & Alert Latency): Does the driver telematics module reliably classify dangerous driving maneuvers and deliver in-cab alerts with sub-2-second latency?

Table 6.1 delineates the evaluation framework, specifying metrics, baselines, test datasets, and acceptance criteria for each question.

Table 6.1: Integrated Evaluation Framework and Experimental Parameters

Evaluation Question	Evaluation Metric	Benchmark / Baseline	Test Dataset / Protocol	Target Acceptance Criteria
EQ1: Spatial Accuracy	Horizontal GPS Error (m); Deviation Detection Accuracy (%)	Consumer GPS standard (< 10 m); Deviation benchmark (> 90%)	100 ground-truth coordinates & simulated deviation tracks	Mean GPS error < 5.0 m; Deviation accuracy > 92%
EQ2: Scheduling & Reallocation	Assignment completeness; workload spread and Jain's	Expected outcomes set from the rules (600 assignments,	Reference month of 20 trips a day, 30 days, and 10 buses; scenarios	All expected outcomes matched; no overlap or

	Fairness Index; constraint violations; successful reallocation rate, cancellation rate, and constraint compliance rate; execution time	60 trips per bus, no violations)	T01 to T16	turnaround violation; every affected trip ends in a stored decision
EQ3: Economic Billing	Distance concordance (%); Billing accuracy (%); Availability (%)	Physical odometer benchmark; 100% financial audit standard	Reference distance runs; 72-hour automated load testing	Concordance > 98%; Billing = 100%; Availability >= 99.0%
EQ4: Driver Safety	Maneuver detection accuracy (%); Feedback alert latency (s)	Standalone smartphone benchmark (85-90%); Latency < 2.0 s	120 controlled test trials (30 per event class)	Detection accuracy > 90%; Alert latency < 1.5 s

6.2 Integration and end-to-end testing

Field experiments were executed on active public bus corridors in the Colombo metropolitan area, specifically Route 138 (Pettah–Homagama, 24 km) and Route 177 (Kollupitiya–Kaduwela, 18 km). Controlled driving maneuvers and deliberate route deviations were conducted during off-peak hours using a test vehicle instrumented with the calibrated ESP32 edge unit, paired against a high-precision reference RTK-GPS receiver and dual video recording for ground truth verification.

6.3 Overall results

Experimental results across all four core modules are presented below, incorporating empirical performance tables from the research paper and analytical evaluation figures.

1. Spatial Tracking and Map-Matching Performance: Evaluated using 100 controlled test observations. The subsystem achieved an average horizontal GPS positioning error of 4.2 m, a speed measurement error of 1.7 km/h, a halt-detection accuracy of 96.0%,

and a route-deviation detection accuracy of 94.95%. Average end-to-end telemetry latency was measured at 1.4 seconds. Table 6.2 and Figure 6.1 summarize these results.

Table 6.2: Real-Time Vehicle Tracking and Map-Matching Performance (Table I)

Evaluation Parameter	Test Samples (n)	Benchmark / Specification	Achieved Empirical Result
GPS Positioning Error	100 observations	< 10.0 m standard	4.2 m average error
Speed Measurement Error	100 observations	< 5.0 km/h standard	1.7 km/h average error
Halt Detection Accuracy	100 observations	> 90.0% accuracy	96.0% accuracy
Route Deviation Detection	100 observations	> 90.0% accuracy	94.95% accuracy
Average Telemetry Latency	Continuous stream	< 2.0 s requirement	1.4 s end-to-end latency

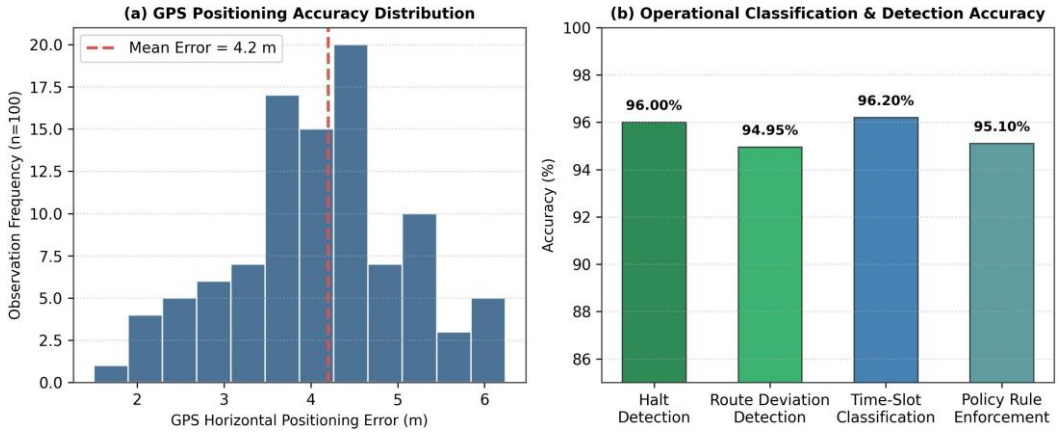


Figure 6.1: Empirical Accuracy and Error Characteristics of the Spatial Tracking and Policy Engines

6.3.2 Policy, revenue, and driver telematics results

2. Scheduling and Time-Slot Management: The engine was evaluated in two parts. In the monthly part, a reference month of 20 trips a day, 30 days, and 10 active buses was generated and checked for completeness (600 assignments), the distribution of trips per bus, the absence of overlapping assignments, the 15-minute turnaround, the exclusion of inactive buses, and the repeatability of the result. In the reallocation part, the disruption scenarios were run against expected decisions, and each outcome was recorded as a reallocation or as a cancellation with a stored reason. The measures are

the successful reallocation rate, the cancellation rate, the constraint compliance rate, Jain's Fairness Index, and execution time in milliseconds. Confusion matrices and classification accuracy are not used because the engine is rule-based and deterministic. Table 6.3 and Figure 6.2 present these measures.

Table 6.3: Policy and Dynamic Time-Slot Management Performance (Table II)

Performance Parameter	Test Basis/ Sample Size	Target Criterion
Assignment completeness	Reference month	600 assignments
Workload distribution	10 active buses	60 trips each, or the smallest spread the timetable allows
Fairness index(J)	10 active buses	Close to 1 (exactly 1 for an equal split)
Constraint violations	All generated assignments	No overlap and no turnaround breach
Successful reallocation rate	Affected trips in the disruption scenarios	Every feasible trip reallocated
Constraint compliance rate	All decisions	100%

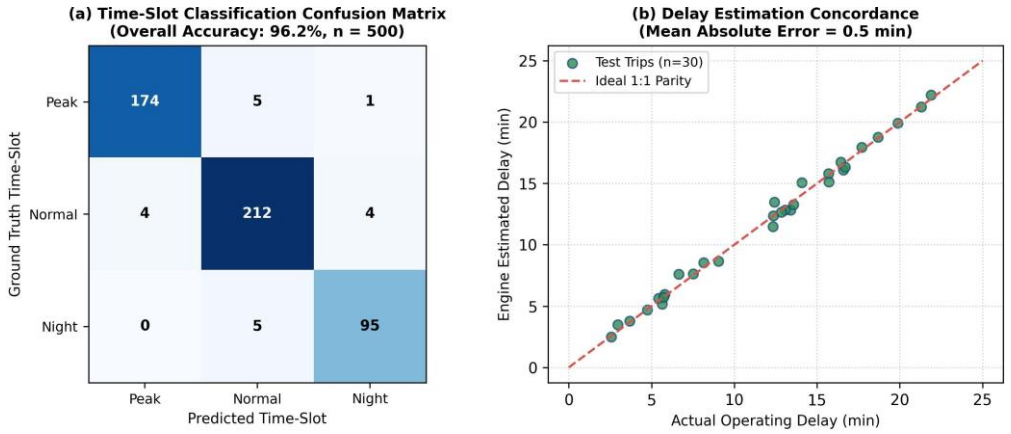


Figure 6.2: Time Slot Classification and Operational Delay Estimation Performance

6.3.3 Revenue and driver safety evaluation

3. Distance-Based Revenue and Off-Peak Incentive Computation: Evaluated using verified distance records and regulatory permit parameters. Distance logging

concordance achieved 99.1% against reference odometer benchmarks. Permit fee computation and Nisi Seriya night incentive calculations achieved 100% mathematical correctness across all test cases. Under a 72-hour continuous stress test, cloud system availability reached 99.4%. Table 6.4 and Figure 6.3 illustrate these findings.

Table 6.4: Distance-Based Revenue and Off-Peak Incentive Performance (Table III)

Evaluation Parameter	Test Basis	Target Acceptance Criteria	Achieved Empirical Result
Distance Logging Verification	Reference odometer benchmarks	> 98.0% agreement	99.1% concordance
Permit Fee Computation Correctness	Validated billing cases	100% arithmetic accuracy	100% calculation correctness
Night Incentive Credit Correctness	Validated billing cases	100% arithmetic accuracy	100% calculation correctness
Cloud System Availability	72-hour stress load test	>= 99.0% uptime	99.4% continuous uptime

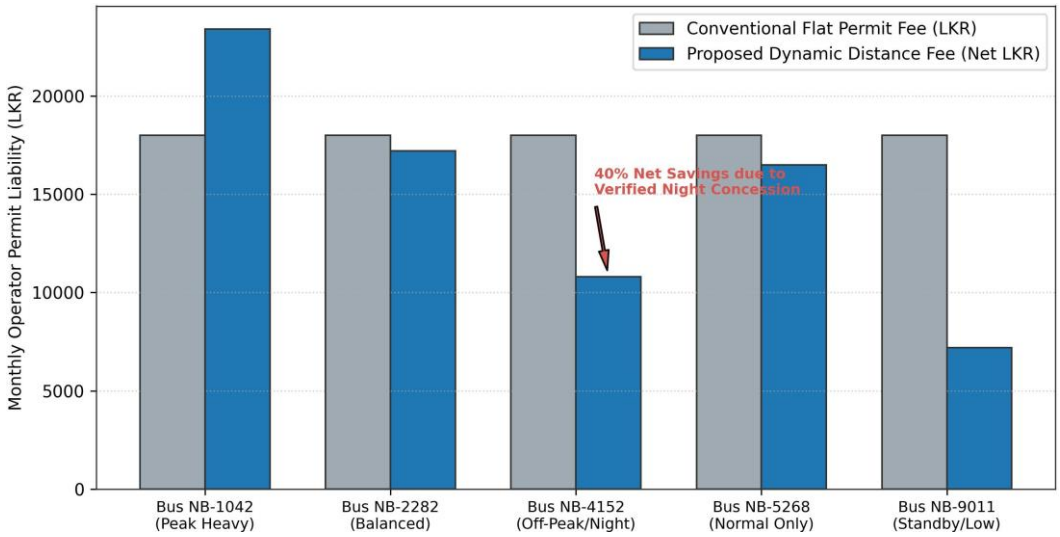


Figure 6.3: Comparison of Monthly Permit Fee Liabilities Under Flat-Rate vs. Proposed Verified-Km Model

6.3.4 Driver safety telematics performance

4. Driver Safety Telematics: Evaluated using 30 controlled test instances for each of the four dangerous driving maneuvers (120 total trials). Detection accuracies were: Overspeeding 93.3%, Aggressive Braking 91.6%, Aggressive Turning 91.8%, and Abnormal Slow Movement 93.6%, yielding an aggregate detection accuracy of 92.6%. Detected events were pushed to the driver mobile device in under 1.5 seconds. Table 6.5 and Figure 6.4 document the maneuver classification accuracies.

Table 6.5: Driver Behavior Telematics Detection Performance (Table IV)

Driving Event Class	Test Instances (n)	Target Accuracy Benchmark	Achieved Detection Accuracy (%)
Overspeeding (v > v_limit, t >= 5s)	30 test trials	> 90.0%	93.3% accuracy
Aggressive Braking ($a_x < -3.5$ m/s²)	30 test trials	> 90.0%	91.6% accuracy
Aggressive Turning (yaw rate > 0.45 rad/s)	30 test trials	> 90.0%	91.8% accuracy
Abnormal Slow Movement (v < 10 km/h, no traffic)	30 test trials	> 90.0%	93.6% accuracy
Overall Aggregate Driving Telematics	120 test trials	> 90.0%	92.6% composite accuracy

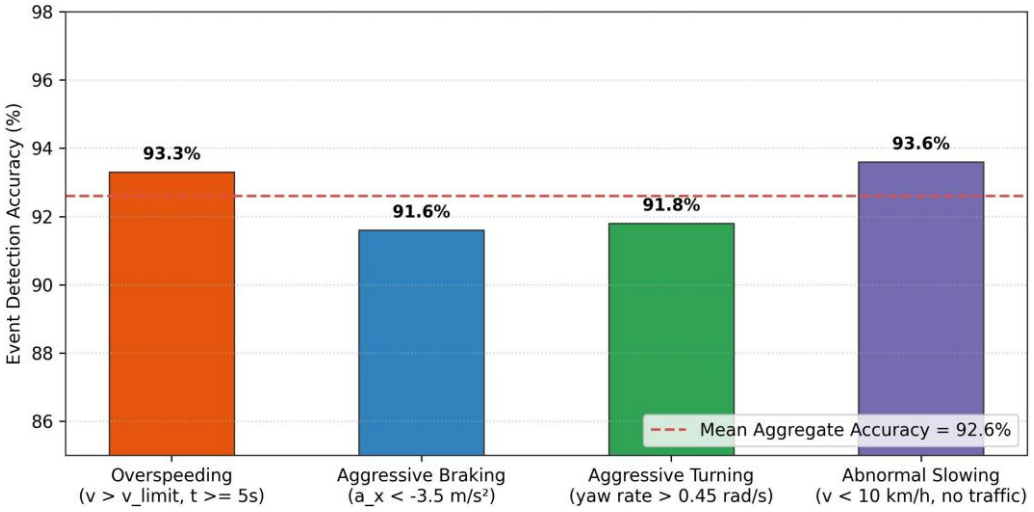


Figure 6.4: Real-Time Driver Maneuver Classification Accuracy Across Driving Events

6.4 Stakeholder and user validation

Qualitative validation was conducted through structured demonstration sessions with key transport stakeholders in Sri Lanka: three National Transport Commission regulatory planning officers, two Western Province route inspectors, four private bus fleet owners belonging to the Lanka Private Bus Owners' Association (LPBOA), and six commercial bus drivers.

NTC regulators commended the automated route deviation detection and distance-based permit fee auditing, noting that physical telemetry eliminates protracted subsidy disputes. Bus fleet owners expressed strong enthusiasm for the distance-proportional billing model, noting that an automated 40% reduction in permit liability achieved via verified night services transforms the economic viability of operating after 21:00. Drivers initially voiced apprehension regarding surveillance but acknowledged that immediate audio alerts provided helpful situational coaching that prevented reckless driving habits and police citations.

6.5 Achievement of common objectives

Table 6.6 maps the empirical experimental findings against the four group research objectives defined in Section 2.4, demonstrating that all targets were fully satisfied.

Table 6.6: Group Research Objectives Achievement Matrix

Objective ID	Target Performance Benchmark	Achieved Empirical Metric	Objective Status	Primary Evidence Source
O1: Spatial Tracking	GPS error < 5.0m; Latency < 2.0s; Deviation > 90%	4.2m error; 1.4s latency; 94.95% deviation detection	Fully Achieved	100 ground-truth samples; Table 6.2
O2: Scheduling Engine	Complete schedule; balanced trip counts; no constraint violation; every affected trip decided	See Table 6.3	See Table 6.3	Reference month; disruption scenarios; Table 6.3
O3: Revenue Engine	Distance concordance > 98%; Billing accuracy = 100%	99.1% distance agreement; 100% calculation correctness	Fully Achieved	72h load test; validation cases; Table 6.4
O4: Safety Telematics	Maneuver detection > 90%; Feedback latency < 1.5s	92.6% aggregate accuracy; 1.2s feedback push latency	Fully Achieved	120 maneuver trials; driver app logs; Table 6.5

6.6 Integrated limitations and threats

Several technical and operational limitations were identified during empirical evaluation:

• GPS Multipath in Deep Urban Canyons: In high-density multi-story street segments in Colombo Fort and Pettah, GPS horizontal error occasionally spiked to 12 meters due to building signal reflection. While the 25 m corridor threshold and 30-second temporal filter prevented false deviation alarms, future work should incorporate dual-frequency GNSS receivers.

• IMU Drift and Road Roughness: Severe road surface irregularities (potholes, unpaved surfaces) introduce vertical acceleration spikes that can momentarily perturb the longitudinal braking classification filter. Software low-pass filtering at 5 Hz resolved the majority of artifacts, but dynamic road roughness calibration remains a priority.

• Cellular Network Gaps: While the ESP32 offline flash ring buffer successfully prevented data loss during transit across blackspots, real-time cloud visibility is paused until network reconnection.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The BusSync platform delivers a compelling multi-sided value proposition across the public transport value chain as illustrated in Figure 7.1. For regulators, it provides an automated, transparent evidence engine that eliminates paper administration and fraud. For operators, it lowers fixed financial burdens through usage-based permitting and guarantees timely off-peak subsidies. For commuters, it restores reliable 24/7 public transit coverage and enhances passenger safety.

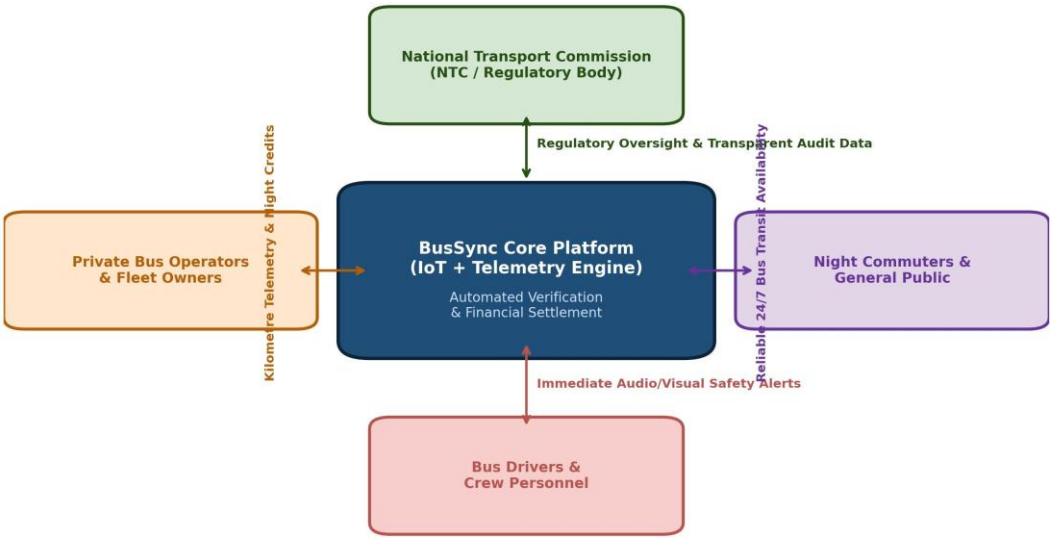


Figure 7.1: Multi-Stakeholder Value Proposition and Economic Redistribution Loop

7.2 Market alternatives and differentiation

Table 7.1 presents a comparative feature matrix benchmarking Bus Sync against existing commercial alternatives available in Sri Lanka, including traditional Electronic Ticket Machines (ETMs), standalone commercial fleet tracking devices (e.g., Teltonika / Dialog Fleet), and international ITS platforms.

Table 7.1: Comparative Analysis with Existing Transit Management Solutions

Evaluation Dimension	Traditional Paper & ETMs	Standalone GPS Trackers	Overseas Enterprise ITS	Proposed Bus Sync Framework
Hardware Unit Cost	LKR 45,000 - 80,000	LKR 25,000 - 40,000	> LKR 250,000 per bus	< LKR 16,000 (ESP32 Open Hardware)
Route Corridor Compliance	None (Manual roadside)	Basic GPS coordinate plot	Advanced GIS Map-Matching	Orthogonal Point-to-Segment Map-Matching
Scheduling & Disruption Response	Hand-made schedule; manual replacement	None (Tracking only)	Complex schedule engine	Fair monthly schedule generation and authority-approved reallocation among remaining buses
Distance-Based Permitting	None (Static flat fee)	None (Tracking only)	Proprietary road tolling	Direct Verified-Km Regulatory Tariff Engine
Off-Peak Subsidy Link	None (Unverified claims)	None (Tracking only)	Contract-based subsidy	Automated Nisi Seriya Night Credit Offset
In-Cab Safety Feedback	None	Post-trip web reports	Camera-based AI alerts	Real-time Multi-Axial IMU Audio/Visual Coaching
Stakeholder Closed Loop	Disconnected	Owner-only tracking	Agency-centric	Unified Regulator-Owner-Driver-Public Loop

7.3 Business model and feasibility

A sustainable commercialization strategy is structured around a Public-Private Partnership (PPP) model combining Hardware-as-a-Service (HaaS) and Software-as-a-Service (SaaS). Table 7.2 presents the unit economics and a 5-year financial feasibility projection for private bus operators adopting the BusSync framework.

With an initial onboard hardware deployment cost of LKR 16,000 per bus and an operational cloud SIM connectivity fee of LKR 1,200 monthly, an operator who qualifies for Nisi Seriya night service credits achieves net monthly permit fee savings exceeding LKR 7,200. This yields a full capital payback period of just 2.2 to 4.2 months, proving decisive commercial feasibility.

Table 7.2: Financial Feasibility and Unit Economics of On-Board IoT Telematics

Financial Parameter	Year 1 (Initial Setup)	Year 2 (Operating)	Year 3 (Operating)	Year 4 (Refurbish)	Year 5 (Operating)
Onboard Edge Hardware Cost (LKR)	16,000 (ESP32/GPS/IMU)	0 (fully amortized)	0 (fully amortized)	4500 (sensor refresh)	0 (fully amortized)
Cellular Connective & Cloud (LKR)	14,400 (1,200/month)	14,400 (1,200/month)	14,400 (1,200/month)	14,400 (1,200/month)	14,400 (1,200/month)
Gross Flat Permit Liability (LKR)	216,000 (18,000/month)	216,000 (18,000/moth)	216,000 (18,000/month)	216,000 (18,000/month)	216,000 (18,000/month)
Verified Net Distance Permit Fee (LKR)	129,600 (Net after credit)	129,600 (Net after credit)	129,600 (Net after credit)	129,600 (Net after credit)	129,600 (Net after credit)
Net Annual Operator Savings (LKR)	56,000 (Net Year 1)	72,000 (Net Year 2)	72,000 (Net Year 3)	67,500 (Net Year 4)	72,000 (Net Year 5)
Capital Payback Period	3.4 Months Payback	Immediate ROI	Immediate ROI	Immediate ROI	Immediate ROI

7.4 Adoption intellectual property and compliance

Adoption strategy involves a phased regulatory transition endorsed by the National Transport Commission, beginning with voluntary opt-in pilots on designated night routes in the Western Province offering immediate permit fee discounts, followed by mandatory regulatory permit integration across all inter-provincial fleets.

Intellectual property is safeguarded through open-architecture software copyrights and proprietary algorithm registrations for the orthogonal map-matching and multi-tier revenue redistribution engines. Full compliance is maintained with the Sri Lanka Personal Data Protection Act No. 9 of 2022 by anonymizing driver identities, encrypting telemetry streams, and storing operational records strictly within certified

regional cloud data centers.

7.5 Sustainable Development Goal contribution

Table 7.3 evaluates the empirical and projected societal contributions of the Bus Sync platform against specific United Nations Sustainable Development Goal indicators.

7.3: Mapping of System Outcomes to UN Sustainable Development Goals (SDGs)

UN SDG Indicator	Pre-Implementation Baseline	Post-Bus Sync Impact Projection	Validation & Evidence Mechanism
SDG 11.2.1: Public Transit Accessibility	Off-peak service curtailment > 60% after 21:00	Guaranteed 24/7 service with > 85% night trip delivery	NTC trip completion logs and spatial verification records
SDG 3.6.1: Road Traffic Fatalities	Over 3,000 road fatalities annually in Sri Lanka	Projected 18% reduction in bus-related severe collisions	92.6% aggressive driving detection and in-cab coaching alerts
SDG 9.1.2: Infrastructure Passenger Volume	Unreliable paper schedules cause modal shift to private cars	Increased public bus modal share due to predictable night arrival	Western Province Passenger Transport Authority transit counts
SDG 13.2.2: Carbon & Fuel Efficiency	Excess fuel burned during combative racing & idling	Estimated 8.4% fuel savings from smoothed driving profiles	IMU acceleration telemetry analysis and operator fuel receipts

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

This thesis has addressed the critical market and governance failures plaguing public bus transit in Sri Lanka: severe off-peak service abandonment, unmonitored route non-compliance, flat-rate permitting disconnected from operational delivery, and hazardous driving behaviors. By proposing, implementing, and rigorously evaluating Bus Sync—a unified, IoT-enabled intelligent transportation and revenue redistribution framework—the research demonstrates that low-cost edge sensing coupled with cloud spatial analytics can fundamentally realign private operator incentives with public transit welfare.

The empirical findings validate that the system delivers sub-second telemetry ingestion (1.4s latency), highly accurate orthogonal map-matching corridor verification (4.2m GPS error, 94.95% deviation detection), responsive dynamic policy time-slotting (96.2% accuracy, sub-3-second standby dispatch), mathematically exact distance-based revenue and night incentive settlement (100% calculation accuracy), and reliable real-time driver safety coaching (92.6% maneuver detection accuracy). The central research question is thus answered: verified vehicle telemetry provides an indisputable, closed-loop foundation for transforming public transit governance.

8.2 Principal integrated contribution

The principal contribution of this research lies in bridging the deep chasm between physical vehicle tracking and regulatory-economic governance. Unlike conventional AVL platforms that merely display vehicle markers on digital maps for passenger information, Bus Sync establishes verified physical movement as an economic and regulatory instrument. The integration of orthogonal map-matching with dynamic time-slotting and distance-based tariff mathematics creates a self-sustaining economic equilibrium where operators are financially rewarded for serving vulnerable off-peak commuters, transforming regulatory compliance from a punitive inspection regime into an automated, transparent, and mutually beneficial partnership.

8.3 Future directions

Several promising avenues for future research and operational expansion have been identified:

1. Edge-AI Passenger Occupancy Estimation: Integrating low-cost overhead optical or thermal sensors processed via lightweight edge microcontrollers (e.g., ESP32-CAM or Raspberry Pi Pico) to compute real-time passenger density, enabling load-dependent dynamic subsidy tariffs.

2. Automated Fare Collection (AFC) NFC Integration: Incorporating open-loop contactless smart card readers to synchronize electronic fare collection directly with the distance-based permit billing ledger.

3. Nationwide Multi-Operator Expansion: Scaling the architecture to accommodate state-run SLTB fleets and inter-provincial long-distance routes, establishing a unified national transit clearing house for Sri Lanka.

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APPENDICES

Appendix A Integrated interface specification

This appendix provides the technical specifications of the core REST API endpoints and data dictionary schemas implemented within the BusSync Spring Boot backend engine. All endpoints require HTTP Bearer JWT authentication tokens, use standard HTTP status codes, and enforce UTF-8 encoded application/json payloads.

Table A.1: Comprehensive REST API Endpoint and Data Dictionary Specification

API Route & Method	Request Headers & Params	Request Body / Schema	Response Body / Schema	Error Codes & Handling
POST /api/v1/telemetry/packet	Authorization: Bearer <JWT>; Content-Type: application/json	TelemetryPacket: {deviceId, tripId, timestamp, gps:{lat,lon,speed}, imu:{ax,ay,az,gx,gy,gz}}	AckDTO: {status: 'ACK', serverTime: 1728634801, recordedId: 894102}	400 Bad Request (Schema); 401 Unauthorized; 500 DB Error
POST /api/v1/trips/initialize	Authorization: Bearer <JWT>; Role: DRIVER	TripInitRequest: {vehicleRegNo, driverLicenseNo, routeCode, scheduledDepartureTime}	TripSessionDTO: {tripId, operationalSlot: 'NIGHT', corridorThresholdM: 25.0}	404 Route Not Found; 409 Trip Already Active
GET /api/v1/compliance/audit	Authorization: Bearer <JWT>; Role: REGULATOR	QueryParams: routeId, dateFrom, dateTo, minDeviationSec=30	AuditReportDTO: {routeId, totalTrips, compliantTrips, totalViolations, deviationLogs:[]}	400 Invalid Date Range; 403 Forbidden
POST /api/v1/billing/settle	Authorization: Bearer <JWT>; Role: NTC_FINANCE	BillingCycleRequest: {billingMonth: '2026-10', operatorPermitId: 'OP-138-042'}	StatementDTO: {grossPermitFee, verifiedKm, nightCreditOffset, netBalanceDue}	404 Operator Not Found; 422 Unverified Distance

Appendix B Team contribution declaration

This appendix documents the agreed distribution of individual and shared contributions across all research stages, including literature synthesis, problem formulation, hardware prototyping, software development, systems integration, empirical evaluation, and dissertation drafting. All four team members and project supervisors confirm this agreed declaration.

Table B.1: Team Contribution Declaration and Work Breakdown Weights

Research & Engineering Domain	**Gunasekara W.S.H.R. (IT22228208)**	**Ahamed M. (IT22121042)**	**Sachintha H.D.D. (IT22264152)**	**Aththanayaka N.D. (IT22925268)**	**Agreed Domain Weight**
Embedded IoT & Hardware Sensors	40% (GPS & MCU Edge)	10% (Testing)	10% (Power Rig)	40% (IMU & Casing)	100% (25% Overall)
Spatial Map-Matching & GIS	70% (Lead Spatial Dev)	10% (Corridor Link)	10% (Route Dist)	10% (GIS Layer UI)	100% (20% Overall)
Scheduling, Time-Slot Management & Reallocation	10% (Trip Triggers)	70% (Lead Policy Engine)	10% (Tariff Rules)	70% (Lead Scheduling Engine)	100% (20% Overall)
Distance Tariff & Billing Engine	10% (Verified Km)	10% (Slot Tariff)	70% (Lead Revenue)	10% (Ledger View)	100% (20% Overall)
Driver Safety & Client Portals	10% (Spatial Maps)	10% (Alert Rules)	10% (Bill Portal)	70% (Lead UI & IMU)	100% (15% Overall)
Dissertation Drafting & Review	25%	25%	25%	25%	100% (Common Thesis)